Evaluating the Impact of Semantic Depth and Domain Adaptation on Medical Question Answering Quality

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Abstract

Medical question answering (QA) systems play a vital role in supporting clinicians, providing information that potentially impacts a patient's treatment and life. However, the quality of such answers can be influenced by a range of technical factors, leading to incorrect decisions in critical scenarios, justified by flawed reasoning. This work aims to address both the calibre of classification decisions and generated rationale behind them, creating an effective QA pipeline. Both proposed classification strategies, Interaction Space LDA and Augmented Cross Encoder showed promise, with accuracy of 72% and macro-average F1 score of 58% respectively. For providing transparent reasoning, Retrieval Augmented LLM Generation was found to significantly outperform a T5 encoder-decoder baseline, with 89% of explanations semantically correct. These experiments help to justify continued research, optimising medical AI systems that better aid healthy human life.

1 Introduction

1.1 Task Background and Motivation

As authoritative sources of medical information have become widely available, QA systems have emerged, aiming to provide quick, clear answers to real medical questions. Clinicians no longer have to trawl through mountains of publications, with AI systems instead carrying the burden. Such systems must handle the complex format, jargon and criticality of medical information, while protecting patient anonymity, a difficult task. This work [1] is motivated by the need to foster trust between clinicians and AI systems, managing ambiguity and justifying answers with explicit medical reasoning.

1.2 PubMedQA Dataset

PubMedQA (Jin et al., 2019) is a dataset used to examine how NLP models adapt to the medical domain. It includes 273.5k questions that can be answered as yes/no/maybe or with a longer explanation. The data is in JSON format, with sets of questions, medical context (e.g: background, methods, results) and corresponding short and long answers, as shown in Figure 1.


```
"QUESTION": "Does ischemic preconditioning req
"CONTEXTS": [
  "Ischemic preconditioning (IP) is initiated
  "Experiments were performed an 27 blood-perf
  "The infarct area in control hearts was 6.7+
],
"LABELS": ["BACKGROUND", "METHOD", "RESULTS"],
"YEAR": "2000",
"final_decision": "no",
"LONG_ANSWER": "The preconditioning effect (in
```


Figure 1: PubMedQA Data Structure

This data facilitates two tasks, classification and explanation generation. Classification has two difficulty levels with regards to information visibility, reasoning-required and reasoning-free, making predictions based on just the context or with the rich long answer as help. They are effectively closed and open book versions of PubMedQA. Generation itself is the ultimate reasoning-required task, as the model must reason and produce a logically sound explanation from just the context. This project is completed reasoning required.

1.3 Data Exploration

PubMedQA includes three datasets, with 1k expert annotated, 61.2k unlabeled, and 211.3k artificially generated QA instances each. The large artificial dataset, however, only contains labels 'yes' and 'no' for the final decision, deduced by a heuristic on respective long answers. When considering the artificial and labeled datasets, a class imbalance can be observed (Figure 2), with significantly more decisions labeled 'yes' and only 11% of the labeled set (110 total questions) classed as 'maybe'.

[1] GitHub Repository: https://github.com/joshdfeather/ai-and-text-analytics-team-1

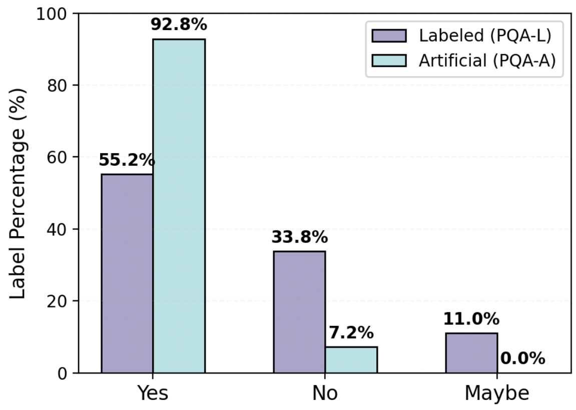

Figure 2: PubMedQA Class Imbalances

The datasets exhibit a diverse range of question types, reasoning styles and statistical interpretations, summarised by Table 1. For example, questions may examine relationships between a factor and an output, or the validity of a statement. Reasoning may compare across groups or examine statistics within them and results may be numerical, using text interpretations or both.

Question Type	**%**
Does a factor influence the output?	36.5
Is a therapy good/necessary?	26.0
Is a statement true?	18.0
Is a factor related to the output?	18.0
Reasoning Type	**%**
Inter-group comparison	57.5
Interpreting subgroup statistics	16.5
Interpreting (single) group statistics	16.0
Text Interpretations of Numbers	**%**
Existing interpretations of numbers	75.5
No interpretations (numbers only)	21.0
No numbers (texts only)	3.5

Table 1: Dataset Diversity (Jin et al., 2019)

The long answers are complicated justifications, acting as a bridge between the categorical decison and relevant context. They average over 40 words, reflecting the complexity of the medical scenario's they describe, with average lengths in Table 2.

Statistic	**PQA-L**	**PQA-U**	**PQA-A**
Avg. long answer length	43.2	45.9	41.0

Table 2: Average Long Answer Lengths

1.4 PubMedQA Solution Requirements

Firstly, a robust approach must include a method quantitative reasoning, involved in 96.5% of results and fundamental to draw conclusions. Secondly, the expert labeled dataset alone does not contain sufficient samples to train a powerful model, so multi-stage fine-tuning with at least 1 more of the datasets is essential. Finally, an effective generation approach must define a clear evaluation strategy to compare explanations to the ground truth long answers, ideally to give a numerical metric.

2 Methods

2.1 Dataset Split and Cross Validation

PubMedQA defines the official data split used to grade leaderboard submissions, which this work copies, partitioning the labeled dataset into a development and holdout testing subset, each with 500 samples. To combat any randomness in the question styles and results, all classification will use 10-fold cross validation and form an ensemble whose predictions are averaged (Figure 3).

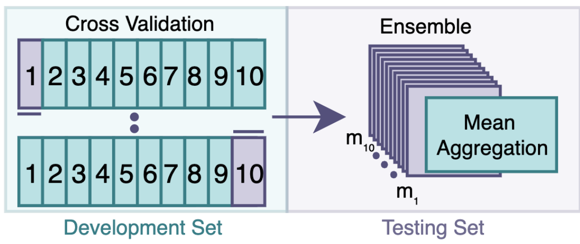

Figure 3: Cross Validation Strategy

2.2 Baseline System

A two stage baseline was initially designed, simply handling classification, predicting short-answers for the labeled dataset, demonstrated by Figure 4.

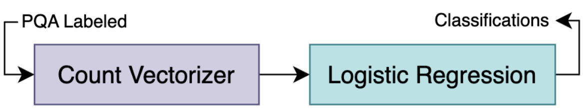

Figure 4: Baseline Pipeline

The question and context are first concatenated into a unified text field. to be passed through a CountVectorizer. This creates a bag of words (BoW) representation, storing the count of every token present. A Logistic Regression Classifier uses the counts as features, attempting to find a decision boundary between them. Initial examination found this over predicted the majority class 'yes' and misclassified all 'maybe' cases (Figure 5).

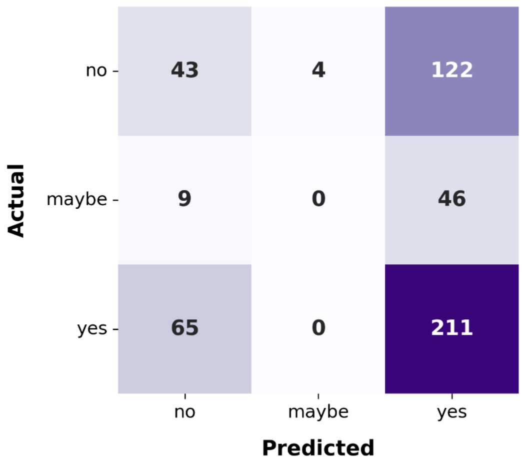

Figure 5: Spare vs Dense Representation

> **Note:** for the later introduced Axis 2, a basic LLM is used for the explanation baseline

2.3 Text Representation

Suspicion was raised that simple BoW representations may not detect genuine relationships between context and decisions, failing to draw decision boundaries between the complex classes. To confirm this, a more advanced BoW option was explored, in TF-IDF. This extends the CountVectorizer by accounting for term frequency (how often a token appears in the text) and inverse document frequency (how often a token appears across the entire dataset). It was theorised this could strengthen rare clinical signals within text and reduce common noise. Equations 1-3 below demonstrate TF-IDF (Salton and Buckley, 1988), for term *t*, frequency function *f*, document *d* and corpus *D* size *N*.

$$tf(t,d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}} \tag{1}$$

$$idf(t,D) = \ln\left(\frac{N}{|\{d \in D : t \in d\}|}\right) \tag{2}$$

$$tfidf(t,d,D) = tf(t,d) \cdot idf(t,D) \tag{3}$$

Experiments were carried out comparing this approach to the baseline, using accuracy and macro-f1 score metrics. The justification behind this choice of metrics is further explore within Section 3.1. TF-IDF was also tested for the non-default token size of two (called bi-grams), incorporating the 'N-gram' method. It was identified that all three BoW models performed notably poorly, demonstrated by Table 3. While these methods are good at capturing keywords, they fundamentally disrupt the local word order, treating medical phrases as disconnected word blocks. This lead to the conclusion that sparse representations are insufficient to understand complex clinical uncertainties, justifying their exclusion for future experiments.

Model	**Accuracy**	**Macro-Avg F1**
Count Vectorizer + Log Reg	0.54	0.34
TF-IDF + Log Reg	0.51	0.36
TF-IDF + N-Grams + Log Reg	0.51	0.32

Table 3: BoW Results

Within the official PubMedQA paper (Jin et al., 2019), the best approach identified was an adapted version of BioBERT (Lee et al., 2020). This is a medical dense representation method, that captures interactions between tokens, moving beyond simply the count of them. It is hypothesised that by mapping the text to a high dimensional vector, models can identify nuance and relationships within the clinical information. The shift from simple sparse vectors to dense embeddings is shown in Figure 6, with BERT models the focus of all remaining classification work below.

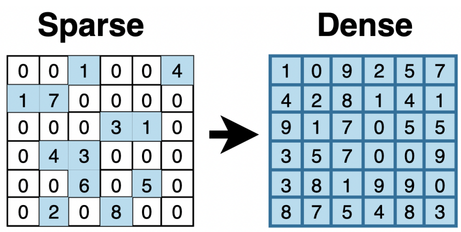

Figure 6: Baseline Confusion Matrix

2.4 Experimental Axes and Approaches

Two axes of experimentation were selected, to evaluate how design choices can impact QA systems and be broken into individual hypotheses.

Axis 1: Quantifying the impact of semantic depth and understanding within medical classification models.

Medical reasoning often hinges on subtle language cues, which effective models must be able to understand. It is expected that moving from simple models, aiming to establish decision boundaries, to token level transformer approaches will allow the pipeline to better understand what distinguishes the classes. Having explored the lowest hierarchy of semantic depth (n-grams), two further levels of understanding are examined, both using embeddings.

2.4.1 Axis 1 Approach 1: Interaction LDA

It is hypothesised that relationships between the corresponding questions (Q) and context (C) have distinct signatures that act to separate the classification classes. By building Interaction Vectors (Reimers and Gurevych, 2019) and visualising them with reduced dimensions, these signatures may become increasingly visible. This approach can be summarised with the pipeline in Figure 7.

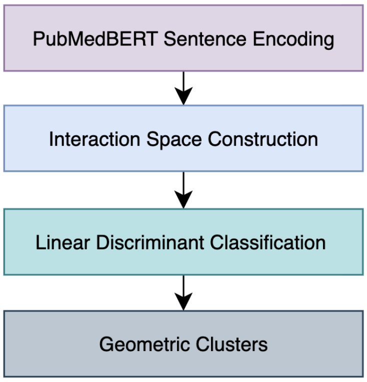

Figure 7: Interaction Space LDA Pipeline

The question and context can first be encoded into seperate embedding vectors $\vec{Q}$ and $\vec{C}$ using a PubMedBERT Sentence Encoder (Gu et al., 2022). In order to compare the resulting embeddings, two compositional functions are applied, with the goal of representing how much the question and context agree and differ. First, to model how closely vectors Q and C overlap, the Hadamard product is used, an element wise multiplication operation that reinforces strong signals present within both vectors. Secondly, to model how the question and context differ, absolute difference is used. The interaction space $\vec{V}$ combines both vectors as in Equation 4.

$$\vec{V} = [|\vec{Q} - \vec{C}|, \vec{Q} \odot \vec{C}] \tag{4}$$

$\vec{V}$ is then partitioned into 10 equally distributed folds, with each training vector normalised using a StandardScaler to ensure any differences caused by the two compositional functions are limited. A Linear Discriminant Analysis Classifier then applies dimensionality reduction to the normalised interaction vectors, aiming to maximise variance between classes. As LDA reduces the interaction space down to just two dimensions (LD1 and LD2), the average ensemble output can be visualised on a 2D scatter plot, hopefully revealing class clusters able to be geometrically separated.

2.4.2 Axis 1 Approach 2: Cross Encoder

It is also conjectured that capturing token level interactions between the question and context, allows for robust and accurate decision making. These represent the largest semantic depth within this project, progressing from simple embedding comparisons. The proposed pipeline in Figure 8 adapts (Jin et al., 2019)'s bootstrapping method.

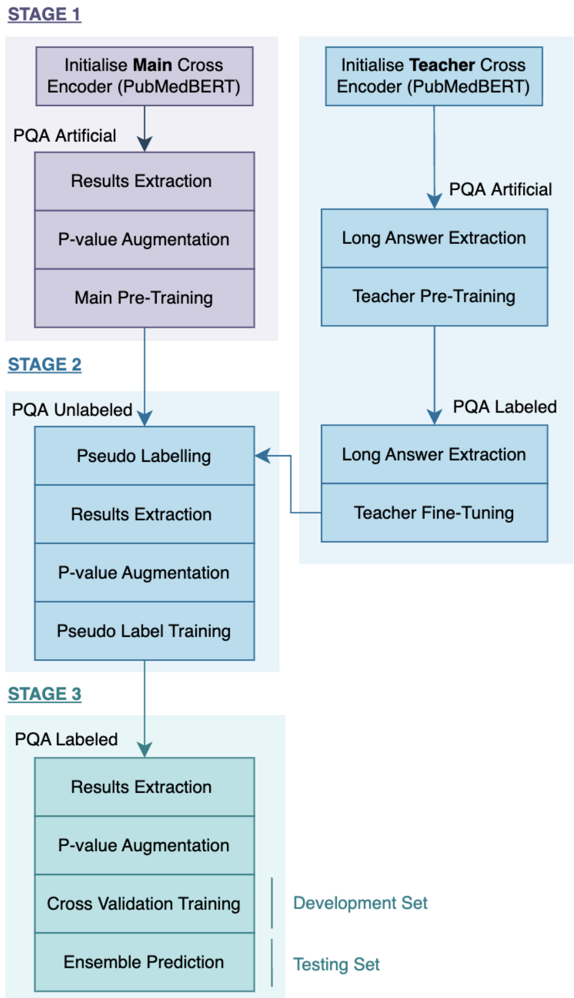

Figure 8: Cross Encoder Pipeline

Stage 1 initialises a Cross Encoder, pre-trained on the artificial data to learn the general structure. Due to Cross Encoder's 512 token sequence limit, processing the whole context is impossible. However, analysis revealed the majority of context contributes limited evidence to aid classification, with the useful findings found within one consistent section, 'results', which is extracted. When inspecting the results, it was found that 43.2% contained P-values (e.g "$p < 0.05$"), a fundamental component to statistical reasoning. They are especially important when deciphering whether differences between two groups are significant (inter-group), 57.5% of PubMedQA's reasoning. As the model treats these identically to any lingusitic token, it may miss their importance, therefore a regex method was designed to help (Figure 9). This performs reasoning for the model, by adding a tag token before the p-value. In research, $p < 0.05$ is the standard used to reject a null hypothesis, therefore these values are tagged as 'significant', and borderline significance is given to results that are interesting but not conclusive.

```
P_REGEX : \b(p(?:-value)?)\s*([<>=])\s*([0-9]*\.?[0-9]+)
FOR PREFIX, OPERATOR, VALUE IN MATCH
IF VALUE <= 0.05
    TAG : "[STATISTICALLY SIGNIFICANT]"
ELIF 0.05 < VALUE <= 0.1
    TAG : "[BORDERLINE SIGNIFICANCE]"
ELSE
    TAG : "[NOT SIGNIFICANT]"
RETURN TAG + PREFIX + OPERATOR + VALUE
```

Figure 9: P-value Regex Pseudocode

Stage 2 implements bootstrapping, to gather additional labeled data. Under reasoning required, long answers are omitted from training, to force reliance purely on signals in the context, however, their clinical rationale can still be leveraged. A separate teacher model, trained with long answers, should be better at ground truth prediction, so can be used to pseudo-label the 61.2k unlabeled samples with hypothesised high accuracy. While these pseudo-samples are used for further tuning, the long answer text itself never directly reaches the main model so the information bottleneck is maintained. Finally, **Stage 3** finishes by fine-tuning on the development partition of labeled data, after extracting results and augmenting p-values.

Axis 2: Evaluating the impact of available knowledge for generating medical explanations

While Axis 1 grows in semantic depth, it only handles categorical decisions, insufficient for real clinicians who require reasoning. Axis 2 acts to explore explanations, comparing the quality of those generated with three approaches: PubMedQA context only (T5), real world knowledge (LLM) and a hybrid approach (RAG), as in Figure 10. LLaMA is used as the LLM baseline.

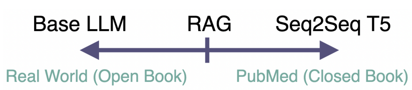

Figure 10: Axis 2 Exploration Summary

2.4.3 Axis 2 Approach 1: Seq2Seq T5

On the PubMedQA reliant end is the Seq2Seq T5 approach. It is hypothesised that this encoder-decoder model could learn a direct mapping from specific PubMedQA context to long answer conclusions, leveraging its ability to summarise. However, due to the limited available data and the task complexity, the small sample size could be prone to hallucinations. The pipeline in Figure 11 will be utilised, following the official testing split.

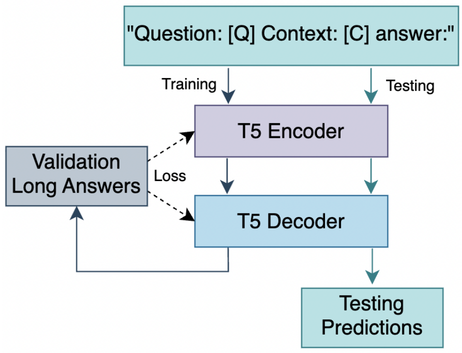

Figure 11: Seq2Seq T5 Pipeline

2.4.4 Axis 2 Approach 2: Retrieval-Augmented Generation

It is hypothesised that Retrieval-Augmented Generation (RAG) can significantly improve the accuracy and reliability of explanations. When answering biomedical questions LLMs may hallucinate, lacking adequate knowledge. However, by retrieving from a grounded knowledge base, the model is provided factual support thought to reduce unsupported inference (Lewis et al., 2021). The pipeline in Figure 12 is further discussed in Section 3.2.2.

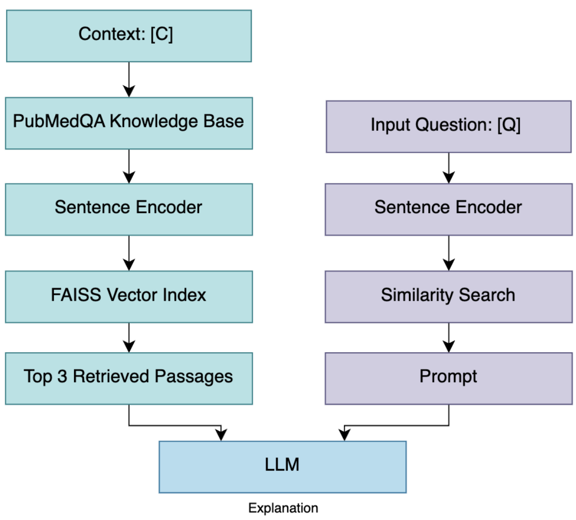

Figure 12: RAG-LLM Pipeline

3 Experimental Evaluation

3.1 Axis 1 Experimentation

To compare classification two metrics are selected, macro-average F1 score and accuracy, demonstrated in Equations 5 - 7 and matching the official benchmark. While accuracy reflects general performance, the class imbalance present means a model could achieve over 55% accuracy by predicting just the majority class (baseline). Consequently, understanding performance across classes can be achieved by pairing macro-F1 score, which balances precision and recall for each class weighted equally. However, a limitation with these metrics is that the cost of misclassification is uniform, unrepresentative of clinical scenarios where certain false positives or negatives have greater impacts. To minimise volatility, StratifiedKFold cross validation was selected (fulfilling Section 2.1), ensuring each fold has consistent quantities of every class.

$$Accuracy = \frac{TP_N + TP_M + TP_Y}{N} \tag{5}$$

$$MacroF1 = \frac{F1_N + F1_M + F1_Y}{3} \tag{6}$$

$$where \quad F1_i = 2 \cdot \frac{P_i \cdot R_i}{P_i + R_i} \tag{7}$$

3.1.1 Interaction LDA Experiment

When building LDA clusters, the Hugging Face model used is PubMedBERT (NeuML, 2026). This is pre-trained on 20,191 PubMed abstracts, learning to map medical text into vector space, perfectly suited to the task within Approach 1. The pipeline in Section 2.4.1 resulted in the clustering displayed through Figure 13, with axes LD1 and LD2 representing the projected coordinate system. Each question within the testing set is plotted as single point, overlaying the 95% confidence intervals for each class. It can be observed that within the reduced interaction space, 'no' and 'yes' are somewhat distinguishable along the first discriminant (LD1), which could represent the clinical polarity, occupying the left and right regions respectively. The maybe cluster appears most separable along the second discriminant (LD2), potentially capturing conflict within evidence.

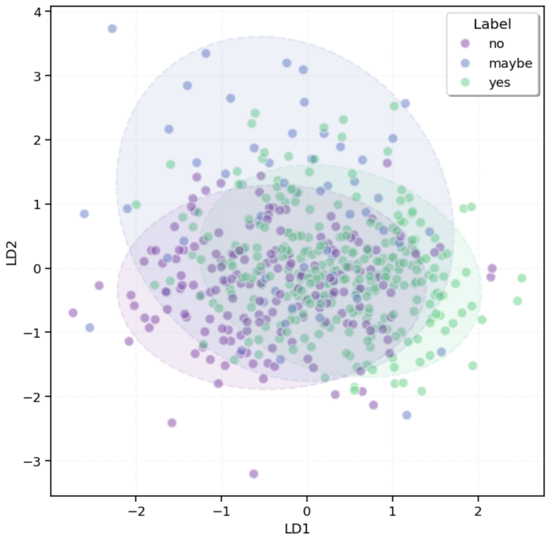

Figure 13: LDA Cluster Diagram

3.1.2 Cross Encoder Experiment

A range of BERT models were explored for the base Cross Encoder, including BioLinkBERT (Michiyasunaga, 2024), DeBerta-V3 (Microsoft, 2021) and Biomed-PubMedBERT (Microsoft, 2025). PubMedBERT was again chosen for the benefits of pre-training in the specific domain. To prevent weights moving quickly in Stages 1 and 2, forgetting PubMedBERT knowledge, label smoothening (Equation 8) was implemented (Müller et al., 2020). Here α ensures capacity is left for future learning by softening the targets y_k for all K classes. Stage 1 used an α of 0.2, avoiding overfitting to the noisy artificial data, decreased in Stage 2 to 0.1. To prevent massive gradient a learning rate (8e-6) was chosen, with deeper learning encouraged just through increased epochs.

$$y'_k = (1 - \alpha) y_k + \frac{\alpha}{K} \tag{8}$$

To combat class imbalance, focal loss was used (Lin et al., 2018), focusing learning on difficult samples (e.g. ambiguous maybe cases). In equation 9 probabilities $1 - p_t$ are raised by γ, the focusing parameter, which punishes low confidence with exponentially greater loss gradients. Following preliminary trials, γ = 1.5 was selected. Class weights were rejected as these encouraged overcompensation, instead over predicting minority classes.

$$FL(p_t) = (1 - p_t)^{\gamma} \log(p_t) \tag{9}$$

The resulting model had strong power to distinguish 'yes' and 'no' classes, and some ability to decipher nuanced 'maybe' cases (Figure 14).

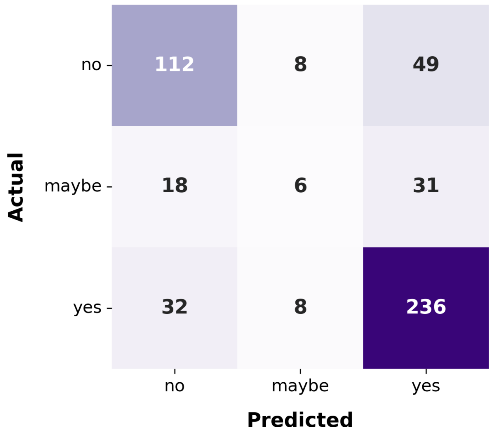

Figure 14: Cross Encoder Confusion Matrix

3.1.3 Classification Failure Case Study

To explore failure modes, a particularly difficult question that both approaches fail on is examined in Figure 15, with true label ‘maybe’.

Question	Expression of c-kit protooncogen in hepatitis B virus-induced chronic hepatitis, cirrhosis and HCC: has it a diagnostic role?
Context	In cirrhotic liver c-kit positivity was rarely present. In the areas where fibrosis was seen, c-kit positivity was rare or absent. The greatest c-kit positivity was found in the livers of patients with severe hepatitis and HCC, in 62 of 75 HCC tissue (p<0.001).

Figure 15: Ambiguous Context

This question is broad, with strong polarised flags, such as c-kit "rarely present" in two thirds of cases but very visible in the third. The confidence for this question from both LDA and the Cross Encoder are plotted in Figure 16. For LDA, a potential limitation revealed here is the issue of softmax labeling, picking the class by purely the highest probability, as here the overlap between binary signals should suggest ambiguity. A Cross Encoder flaw may be identified too, being blinded by the significant p-value and positive section.

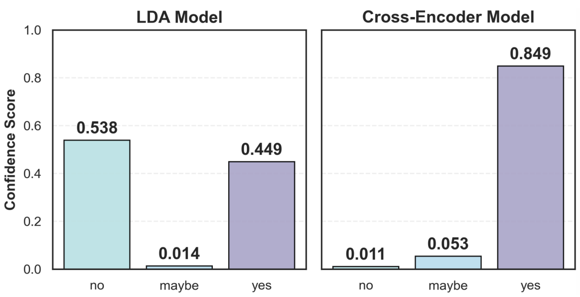

Figure 16: Ambiguous Context

3.1.4 Axis 1 Metrics and Discussion

Results are displayed in Table 4, comparing the Baseline model, LDA and each progressive stage of the Cross Encoder to the Official PubMedQA model. The best results for each metric are **bolded**. When considering accuracy, the Cross Encoder outperformed all alternatives with **0.72**, able to understand clinical polarity for the majority ‘yes’ and ‘no’ classes, aided by p-value augmentation. Each stage served its desired purpose, increasing both the total correct predictions and 2balance across the classes. As LDA is forced to find a single geometric center of each class, it lacks the raw distinguishing capacity of the Cross Encoder, hence its lower accuracy. However, due to this simplicity it generalises better, perhaps less distracted by binary signals, with the highest macro-F1 score of **0.58**.

Model	Accuracy	Macro-Avg F1
Official Model	0.68	0.53
Baseline	0.54	0.34
LDA	0.66	**0.58**
Cross-Encoder		
Stage 1	0.66	0.46
Stage 1-2	0.67	0.48
Stage 1-3	**0.72**	0.54

Table 4: Axis 1 Performance Comparison

3.2 Axis 2 Experimentation

To evaluate explanations against the ground truth, a measure of semantic correctness is required. Natural Language Inference (NLI) (Bowman et al., 2015) was chosen, a task where text assumed to be 100% factual (ground truth) is compared to a hypothesis (explanation). Depending on its confidence in the hypothesis, NLI outputs one of three labels by default: entail, contradict and neutral. CrossEncoder NLI was chosen, able to leverage deep links between text, with DeBERTA-v3 (Microsoft, 2021) selected for its strength judging logical consistency. NLI is configured conservatively, with the explanation only correct in the scenario that the most confident prediction is entailment. Otherwise, to model medical safety, the generated explanation is denoted as unverified, even if NLI decides neutrality. Generation accuracy is therefore the percentage of correct explanations. The pipeline for this system is demonstrated in Figure 17, resulting in an accuracy metric for each model.

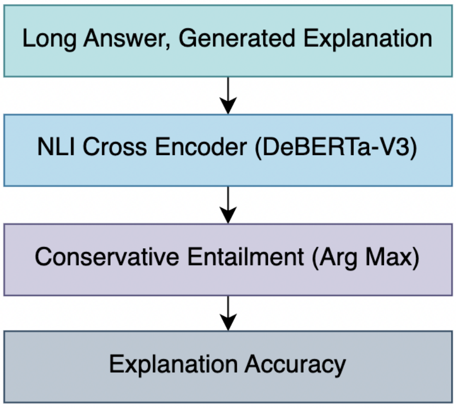

Figure 17: Explanation Evaluation Pipeline

NLI was validated using 90 mock explanations, of which 11% were explicitly incorrect. NLI successfully identified 100% of the incorrect explanations as contradictions and cautiously predicted neutral for 40% of correct cases, as expected.

3.2.1 T5 experiment

Here, a t5-small instance is prompted with each question and context, trained with AdamW (learning rate = 3e-4, batch size = 4, epochs = 1).

3.2.2 RAG-LLM experiment

For RAG, a knowledge base is constructed from context fields across all entries in the PubMedQA labeled dataset, each concatenated and encoded into dense vector representations, using an all-MiniLM-L6-v2 Sentence Transformer (Reimers and Gurevych, 2019). Vectors are stored in a FAISS flat L2 index, serving as the retrieval knowledge base and enabling nearest neighbour search (Johnson et al., 2017). Each test question Q is also encoded into the same dense vector space. The top 3 most similar passages are retrieved from the knowledge base, grounding the LLM with real evidence, combined with the question to form a prompt. The model is instructed to generate explanations grounded solely on the retrieved passages, within an 100 word limit. A lightweight LLM was chosen, using LLaMA 3.2(AI, 2024), an open-source language model deployed locally via Ollama, without API dependency or rate limiting.

3.2.3 Explanation Failure Case Study

Again, to explore generation failures, the quality of explanations for the question in Figure 18 is examined, compared to the true answer.

Question	Blunt trauma in intoxicated patients: is abdomen CT scans always necessary?
Answer	The incidence of abdominal injury in intoxicated blunt trauma patients with normal abdominal examination is low, allowing accurate evaluation without CT
T5	A significant association between abdominal injury and abdominal injury was not necessary in alcohol intoxicated trauma
LLM	Computed CT scans are often necessary to reveal injuries that go undetected. In this study CTs were negative in 98% of cases
RAG	No, computed tomography (CT) of the abdomen is not always necessary and may be unreliable in intoxicated patients. Relying solely on CT scans can reduce unnecessary surgeries for intoxicated patients

Figure 18: Explanation Failure Case

The long answer states that stable patients with a normal abdominal exam, have low incidence of abdominal injury, not requiring CTs. The T5 explanation fails to capture any core arguments, repeating "abdominal injury" in its answer. While coherent, the LLM misses the true reasoning, hallucinating that CT scans are often necessary despite their negativity 98% of the time. Finally, RAG gives a correct interpretation, but confusingly adds an additional statement regarding reduced surgeries, carrying noise from retrieved context. This case acts to demonstrate respective limitations, confirming that while RAG is most coherent and correct, irrelevant context can still cause interference.

3.2.4 Axis 2 Metrics and Discussion

Table 5 reports the final explanation accuracy for all three models with a clear ranking.

Model	Explanation Accuracy
T5	0.72
LLM	0.88
RAG	**0.89**

Table 5: Axis 2 Performance Comparison

The efficient T5 relies solely on limited context extracts, encoding domain information. While useful as a concise baseline, its limited parameters and available information fundamentally restricts it. The LLM, by contrast, draws just upon knowledge from pre-training, enabling fluency without specific evidence, marginally below context retrieval. RAG acts as a middle ground, supplementing real world knowledge with concrete examples from studies, producing the most reliable outputs.

4 Conclusions

This study investigated the impact of text analytics decisions on medical question answering, considering choices at each stage of the QA pipeline.

4.1 Key Classification Findings

As per the hypothesis, semantic depth was confirmed to help discerning between classes. Basic sparse options, even implementing TF-IDF, lacked capacity to capture sentences, uncertainty and statistics, therefore deemed insufficient, with maximum accuracy 54%. Deeper learning was needed to capture underlying relationships within medical text. The interpretable Interaction Space LDA managed class imbalances best, deducing difference between dense vectors, with macro-f1 score 0.58. Conversely, considering raw processing power, the Cross Encoder could reason with individual tokens from the medical extracts, clearly understanding polarised flags. By leveraging pseudo-labelling and p-value augmentation, the model could reason statistically, with accuracy 72%.

4.2 Key Explanation Findings

Axis 2 confirmed that the quality of generated explanations are highly dependent on the training knowledge available. The PubMedQA grounded T5 baseline was found to lack lingusitc competency (72% accurate), while the coherent real-world LLM occasionally made unsupported biomedical claims (88% accurate). The RAG middle ground worked best (89% accurate), combining generative capabilities with medical evidence. This teaches that effective model training should match the education journey of a domain expert, like a human doctor taught to broadly understand the world but grow deep knowledge of medicine.

4.3 Future Work

Throughout this work, the nuances of PubMedQA were understood, identifying the skewed maybe class as the principal challenge in this difficult benchmark. Modelling uncertainty well is a complicated task, not wavering in the presence of binary flags and remaining confident in ambiguity. Calibrating models to best navigate this is the main open challenge for medical QA, with the goal of matching the expertise of a trained professional. LLM explanations also must move towards verifiability, reducing hallucinated, noisy claims that could justify decisions and increasing clinical trust.

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